

Commerce XI Maths App

Exercise 4.1

Detailed Answers with Explanations and Steps

Mathematics & Statistics, Part 2, Unit 4
Bivariate Frequency Distribution and Chi Square Statistic
Total questions: 15

Question 1 - MCQ · PG-49 · EX-Q1(i)

Following table gives income (X) and expenditure (Y) of 25 families:

Y \ X	200-300	300-400	400-500
200-300			
300-400	-		
400-500	-	-	

Find marginal frequency distributions of income and expenditure.

- A. Income: 6, 10, 9; Expenditure: 13, 10, 2
- B. Income: 13, 10, 2; Expenditure: 6, 10, 9
- C. Income: 6, 9, 10; Expenditure: 13, 9, 3
- D. Income: 25, 25, 25; Expenditure: 25, 25, 25

Correct answer: Income: 6, 10, 9; Expenditure: 13, 10, 2

Explanation and steps

The column totals give income frequencies 6, 10, 9 and the row totals give expenditure frequencies 13, 10, 2.

Question 2 - MCQ · PG-49 · EX-Q1(ii)

Following table gives income (X) and expenditure (Y) of 25 families:

Y \ X	200-300	300-400	400-500
200-300			
300-400	-		
400-500	-	-	

Find conditional frequency distribution of X when Y is between 300–400.

- A. 0, 4, 6
- B. 6, 6, 1
- C. 0, 0, 2
- D. 6, 10, 9

Correct answer: 0, 4, 6

Explanation and steps

Read the expenditure class 300–400 row.

Question 3 - MCQ · PG-49 · EX-Q1(iii)

Following table gives income (X) and expenditure (Y) of 25 families:

Y \ X	200-300	300-400	400-500
200-300			
300-400	-		
400-500	-	-	

Find conditional frequency distribution of Y when X is between 200–300.

- A. 6, 0, 0
- B. 6, 4, 0
- C. 6, 6, 1
- D. 13, 10, 2

Correct answer: 6, 0, 0

Explanation and steps

Read the income class 200–300 column.

Question 4 - ENTER ANSWER · PG-49 · EX-Q1(iv)

Following table gives income (X) and expenditure (Y) of 25 families:

Y \ X	200-300	300-400	400-500
200-300			
300-400	-		
400-500	-	-	

Find how many families have their income Rs. 300 and more and expenses Rs. 400 and less?

Correct answer: 17

Explanation and steps

Add the cells for income at least 300 and expenditure below 400: $6 + 1 + 4 + 6 = 17$.

Question 5 - MCQ · PG-49 · EX-Q2

Two dice are thrown simultaneously 25 times. The following pairs of observations are obtained.

(2, 3), (2, 5), (5, 5), (4, 5), (6, 4), (3, 2), (5, 2), (4, 1), (2, 5), (6, 1), (3, 1), (3, 3), (4, 3), (4, 5), (2, 5), (3, 4), (2, 5), (3, 4), (2, 5), (4, 3), (5, 2), (4, 5), (4, 3), (2, 3), (4, 1)

Prepare a bivariate frequency distribution table for the above data.

A.

X \ Y	2	3	4	5	6	Total
1	0	1	2	0	1	4
2	0	1	0	2	0	3
3	2	1	3	0	0	6
4	0	2	0	0	1	3
5	5	0	3	1	0	9
Total	7	5	8	3	2	25

B.

X \ Y	2	3	4	5	6	Total
1	0	1	0	2	0	3
2	0	1	2	0	1	4
3	2	1	3	0	0	6
4	0	2	0	0	1	3
5	5	0	3	1	0	9
Total	7	5	8	3	2	25

C.

X \ Y	2	3	4	5	6	Total
1	0	2	1	0	1	4
2	0	0	1	2	0	3
3	2	3	1	0	0	6
4	0	0	2	0	1	3
5	5	3	0	1	0	9
Total	7	8	5	3	2	25

D.

X \ Y	2	3	4	5	6	Total
1	0	0	2	0	5	7

2	1	1	1	2	0	5
3	2	0	3	0	3	8
4	0	2	0	0	1	3
5	1	0	0	1	0	2
Total	4	3	6	3	9	25

Correct answer:

Y \ X	2	3	4	5	6	Total
1	0	1	2	0	1	4
2	0	1	0	2	0	3
3	2	1	3	0	0	6
4	0	2	0	0	1	3
5	5	0	3	1	0	9
Total	7	5	8	3	2	25

Explanation and steps

Tallying the 25 ordered pairs gives the displayed table.

Question 6 - MCQ · PG-49 · EX-Q2

Two dice are thrown simultaneously 25 times. The following pairs of observations are obtained.

(2, 3), (2, 5), (5, 5), (4, 5), (6, 4), (3, 2), (5, 2), (4, 1), (2, 5), (6, 1), (3, 1), (3, 3), (4, 3), (4, 5), (2, 5), (3, 4), (2, 5), (3, 4), (2, 5), (4, 3), (5, 2), (4, 5), (4, 3), (2, 3), (4, 1)

Obtain the marginal frequency distributions of X and Y.

- A. X: 7, 5, 8, 3, 2; Y: 4, 3, 6, 3, 9
- B. X: 4, 3, 6, 3, 9; Y: 7, 5, 8, 3, 2
- C. X: 7, 8, 5, 3, 2; Y: 4, 3, 6, 3, 9
- D. X: 7, 5, 8, 3, 2; Y: 3, 4, 6, 3, 9

Correct answer: X: 7, 5, 8, 3, 2; Y: 4, 3, 6, 3, 9

Explanation and steps

Column totals give the marginal distribution of X: 7, 5, 8, 3, 2. Row totals give the marginal distribution of Y: 4, 3, 6, 3, 9.

Question 7 - MCQ · PG-49 · EX-Q3

Following data gives the age of husbands (X) and age of wives (Y) in years.

X	27	25	28	26	29	27	28	26	25	25	27
Y	21	20	20	21	23	22	20	20	19	19	23
X	26	29	25	27	26	25	28	25	27		
Y	19	23	23	22	21	20	22	23	22		

Construct a bivariate frequency distribution table.

A.

X \ Y	25	26	27	28	29	Total
19	2	1	0	0	0	3
20	2	1	0	2	0	5
21	0	2	1	0	0	3
22	0	0	3	1	0	4
23	2	0	1	0	2	5
Total	6	4	5	3	2	20

B.

X \ Y	2	3	4	5	6	Total
1	0	1	2	0	1	4
2	0	1	0	2	0	3
3	2	1	3	0	0	6
4	0	2	0	0	1	3
5	5	0	3	1	0	9
Total	7	5	8	3	2	25

C.

X \ Y	20–30	30–40	40–50	Total
20–30	2	2	1	5
30–40	3	2	2	7
40–50	2	1	1	4
Total	7	5	4	16

D.

X \ Y	150–154	155–159	160–164	165–169	Total

35-39	3	1	0	0	4
40-44	2	3	0	0	5
45-49	1	2	1	0	4
50-54	1	1	3	2	7
Total	7	7	4	2	20

Correct answer:

X \ Y	25	26	27	28	29	Total
19	2	1	0	0	0	3
20	2	1	0	2	0	5
21	0	2	1	0	0	3
22	0	0	3	1	0	4
23	2	0	1	0	2	5
Total	6	4	5	3	2	20

Explanation and steps

Classifying the 20 husband-wife age pairs gives the displayed bivariate frequency distribution table.

Question 8 - MCQ · PG-49 · EX-Q3

Following data gives the age of husbands (X) and age of wives (Y) in years.

X	27	25	28	26	29	27	28	26	25	25	27
Y	21	20	20	21	23	22	20	20	19	19	23
X	26	29	25	27	26	25	28	25	27		
Y	19	23	23	22	21	20	22	23	22		

Find the marginal distributions.

- A. X: 6, 4, 5, 3, 2; Y: 3, 5, 3, 4, 5
- B. X: 3, 5, 3, 4, 5; Y: 6, 4, 5, 3, 2
- C. X: 6, 5, 4, 3, 2; Y: 3, 5, 3, 4, 5
- D. X: 6, 4, 5, 3, 2; Y: 5, 4, 3, 5, 3

Correct answer: X: 6, 4, 5, 3, 2; Y: 3, 5, 3, 4, 5

Explanation and steps

Column totals give X: 6, 4, 5, 3, 2. Row totals give Y: 3, 5, 3, 4, 5.

Question 9 - MCQ · PG-49 · EX-Q3

Following data gives the age of husbands (X) and age of wives (Y) in years.

X	27	25	28	26	29	27	28	26	25	25	27
Y	21	20	20	21	23	22	20	20	19	19	23
X	26	29	25	27	26	25	28	25	27		
Y	19	23	23	22	21	20	22	23	22		

Find conditional frequency distribution of age of husbands when the age of wife is 23 years.

- A. 2, 0, 1, 0, 2
- B. 0, 2, 1, 0, 2
- C. 2, 1, 0, 0, 2
- D. 6, 4, 5, 3, 2

Correct answer: 2, 0, 1, 0, 2

Explanation and steps

Read the Y = 23 row for husband ages X = 25, 26, 27, 28, 29: 2, 0, 1, 0, 2.

Question 10 - MCQ · PG-49 · EX-Q4

Construct a bivariate frequency distribution table of the marks obtained by students in Statistics (X) and English (Y).

Marks in Statistics (X)	37	20	46	28	35	26	41	48	32	23	20	39	47	33	27	26
Marks in English (Y)	30	32	41	33	29	43	30	21	44	38	47	24	32	31	20	21

Construct a bivariate frequency distribution table for the above data by taking class intervals 20–30, 30–40, etc. for both X and Y.

A.

X Y	20–30	30–40	40–50	Total
20–30	2	2	1	5
30–40	3	2	2	7
40–50	2	1	1	4
Total	7	5	4	16

B.

X Y	25	26	27	28	29	Total
19	2	1	0	0	0	3
20	2	1	0	2	0	5
21	0	2	1	0	0	3
22	0	0	3	1	0	4
23	2	0	1	0	2	5
Total	6	4	5	3	2	20

C.

X Y	2	3	4	5	6	Total
1	0	1	2	0	1	4
2	0	1	0	2	0	3
3	2	1	3	0	0	6
4	0	2	0	0	1	3
5	5	0	3	1	0	9
Total	7	5	8	3	2	25

D.

X Y	200–300	300–400	400–500	Total
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200–300	6	6	1	13
300–400	0	4	6	10
400–500	0	0	2	2
Total	6	10	9	25

Correct answer:

X \ Y	20–30	30–40	40–50	Total
20–30	2	2	1	5
30–40	3	2	2	7
40–50	2	1	1	4
Total	7	5	4	16

Explanation and steps

Classifying the marks into 20–30, 30–40 and 40–50 for both variables gives the displayed table.

Question 11 - MCQ · PG-49 · EX-Q4

Construct a bivariate frequency distribution table of the marks obtained by students in Statistics (X) and English (Y).

Marks in Statistics (X)	37	20	46	28	35	26	41	48	32	23	20	39	47	33	27	26
Marks in English (Y)	30	32	41	33	29	43	30	21	44	38	47	24	32	31	20	21

Find the marginal distributions.

- A. X: 7, 5, 4; Y: 5, 7, 4
- B. X: 5, 7, 4; Y: 7, 5, 4
- C. X: 7, 4, 5; Y: 5, 7, 4
- D. X: 7, 5, 4; Y: 7, 4, 5

Correct answer: X: 7, 5, 4; Y: 5, 7, 4

Explanation and steps

Column totals give X: 7, 5, 4. Row totals give Y: 5, 7, 4.

Question 12 - MCQ · PG-49 · EX-Q4

Construct a bivariate frequency distribution table of the marks obtained by students in Statistics (X) and English (Y).

Marks in Statistics (X)	37	20	46	28	35	26	41	48	32	23	20	39	47	33	27	26
Marks in English (Y)	30	32	41	33	29	43	30	21	44	38	47	24	32	31	20	21

Find conditional frequency distribution of Y when X lies between 30–40.

- A. 2, 2, 1
- B. 3, 2, 2
- C. 2, 3, 2
- D. 7, 5, 4

Correct answer: 2, 2, 1

Explanation and steps

Read the X = 30–40 column for the three Y classes: 2, 2, 1.

Question 13 - MCQ · PG-49 · EX-Q5

Following data gives height in cm (X) and weight in kgs (Y) of 20 boys.

(152, 40), (160, 54), (163, 52), (150, 35), (154, 36), (160, 49), (166, 54), (157, 38), (159, 43), (153, 48), (152, 41), (158, 51), (155, 44), (156, 47), (156, 43), (166, 53), (160, 50), (151, 39), (153, 50), (158, 46)

Prepare a bivariate frequency table taking class intervals 150–154, 155–159...etc. for X and 35–39, 40–44...etc for Y.

A.

X \ Y	150–154	155–159	160–164	165–169	Total
35–39	3	1	0	0	4
40–44	2	3	0	0	5
45–49	1	2	1	0	4
50–54	1	1	3	2	7
Total	7	7	4	2	20

B.

X \ Y	135–145	145–155	155–165	Total
35–45	2	2	0	4
45–55	2	1	1	4
55–65	4	4	4	12
Total	8	7	5	20

C.

X \ Y	20–30	30–40	40–50	Total
20–30	2	2	1	5
30–40	3	2	2	7
40–50	2	1	1	4
Total	7	5	4	16

D.

X \ Y	200–300	300–400	400–500	Total
200–300	6	6	1	13
300–400	0	4	6	10
400–500	0	0	2	2
Total	6	10	9	25

Correct answer:

X Y	150–154	155–159	160–164	165–169	Total
35–39	3	1	0	0	4
40–44	2	3	0	0	5
45–49	1	2	1	0	4
50–54	1	1	3	2	7
Total	7	7	4	2	20

Explanation and steps

Classifying all 20 height-weight pairs gives the displayed bivariate frequency table.

Question 14 - MCQ · PG-49 · EX-Q5(i)

Following data gives height in cm (X) and weight in kgs (Y) of 20 boys.

(152, 40), (160, 54), (163, 52), (150, 35), (154, 36), (160, 49), (166, 54), (157, 38), (159, 43), (153, 48), (152, 41), (158, 51), (155, 44), (156, 47), (156, 43), (166, 53), (160, 50), (151, 39), (153, 50), (158, 46)

Find the marginal frequency distributions.

- A. X: 7, 7, 4, 2; Y: 4, 5, 4, 7
- B. X: 4, 5, 4, 7; Y: 7, 7, 4, 2
- C. X: 7, 4, 7, 2; Y: 4, 5, 4, 7
- D. X: 7, 7, 4, 2; Y: 5, 4, 4, 7

Correct answer: X: 7, 7, 4, 2; Y: 4, 5, 4, 7

Explanation and steps

Column totals give X: 7, 7, 4, 2. Row totals give Y: 4, 5, 4, 7.

Question 15 - MCQ · PG-49 · EX-Q5(ii)

Following data gives height in cm (X) and weight in kgs (Y) of 20 boys.

(152, 40), (160, 54), (163, 52), (150, 35), (154, 36), (160, 49), (166, 54), (157, 38), (159, 43), (153, 48), (152, 41), (158, 51), (155, 44), (156, 47), (156, 43), (166, 53), (160, 50), (151, 39), (153, 50), (158, 46)

Find the conditional frequency distribution of Y when $155 \leq X \leq 159$.

- A. 1, 3, 2, 1
- B. 3, 1, 0, 0
- C. 4, 5, 4, 7
- D. 7, 7, 4, 2

Correct answer: 1, 3, 2, 1

Explanation and steps

Read the 155–159 column of the constructed table: 1, 3, 2, 1.